INTELLIGENT COGNITIVE ALARM PLATFORM

Infosys Springboard internship 7.0

Title : Intelligent Cognitive Alarm Platform

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PROJECT Objective

Project Objective

Intelligent Cognitive Alarm Platform

To develop an AI-powered alarm platform that helps users build consistent wake-up habits by requiring them to complete personalized cognitive challenges before dismissing an alarm.

Key Objectives:

- Create and manage personalized alarms.
- Use cognitive challenges such as **math, logic, memory, and pattern-based problems** to encourage active wake-up.
- Monitor wake-up behavior, snooze patterns, and challenge performance.
- Adapt challenge difficulty based on user behavior.
- Improve wake-up consistency and reduce oversleeping.
- Provide different functionalities for **Users, Wellness Coaches, and Administrators**.

MILESTONE 1

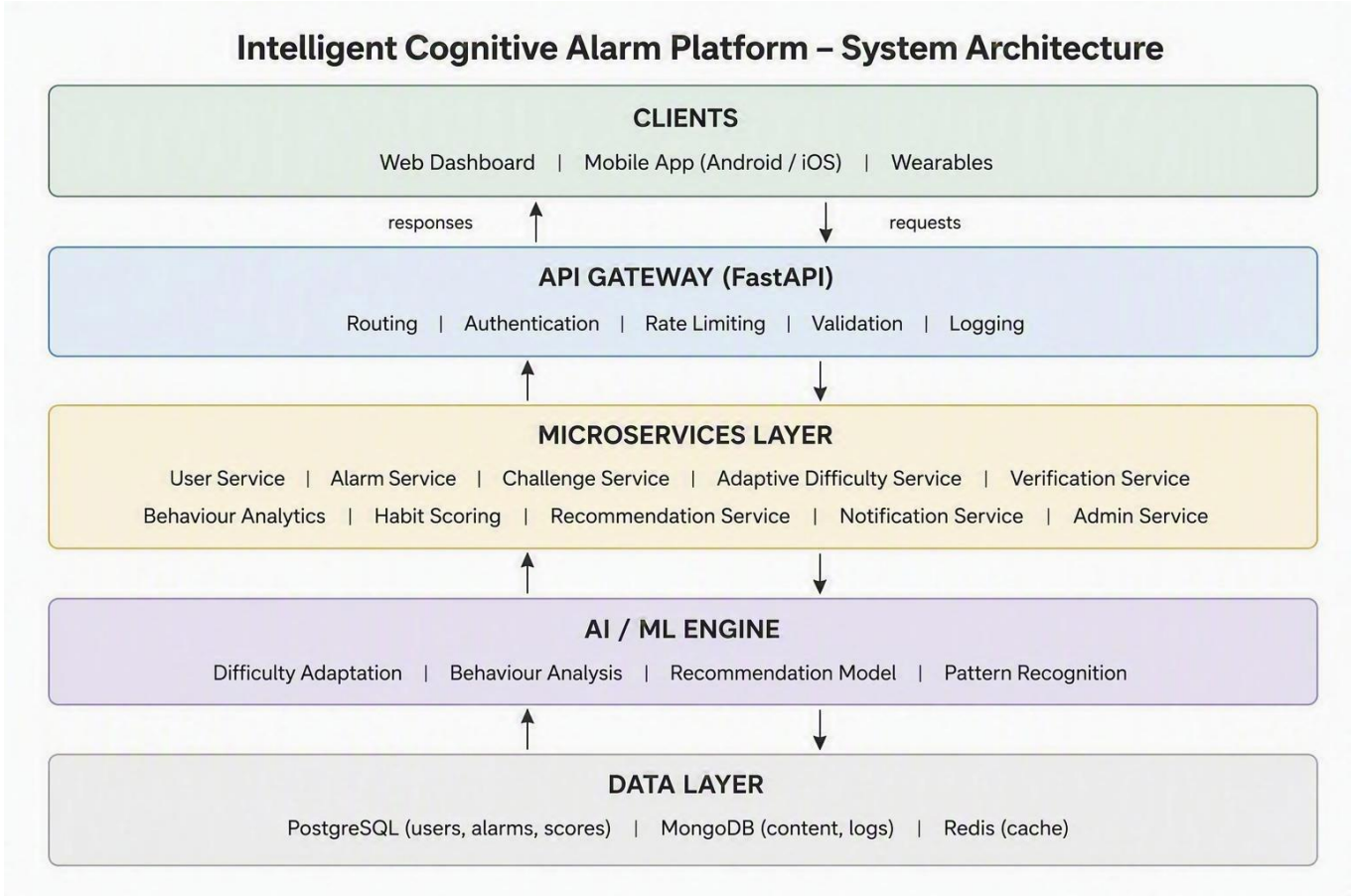
Milestone 1 – Core Foundation

- **Project Initialization** using React.js, FastAPI and SQLite
- **User Authentication** with JWT and secure password hashing
- **Role-Based Access Control** for User, Wellness Coach and Administrator
- **User Profile & Habit Preferences** management
- **Alarm Scheduling** with create, update, delete and enable/disable features
- **Frontend–Backend Integration** through REST APIs
- **Database Design** using SQLAlchemy ORM
- **Outcome:** A functional platform foundation with authentication, user management and alarm management.

SYSTEM ARCHITECTURE

Milestone 1: System Architecture

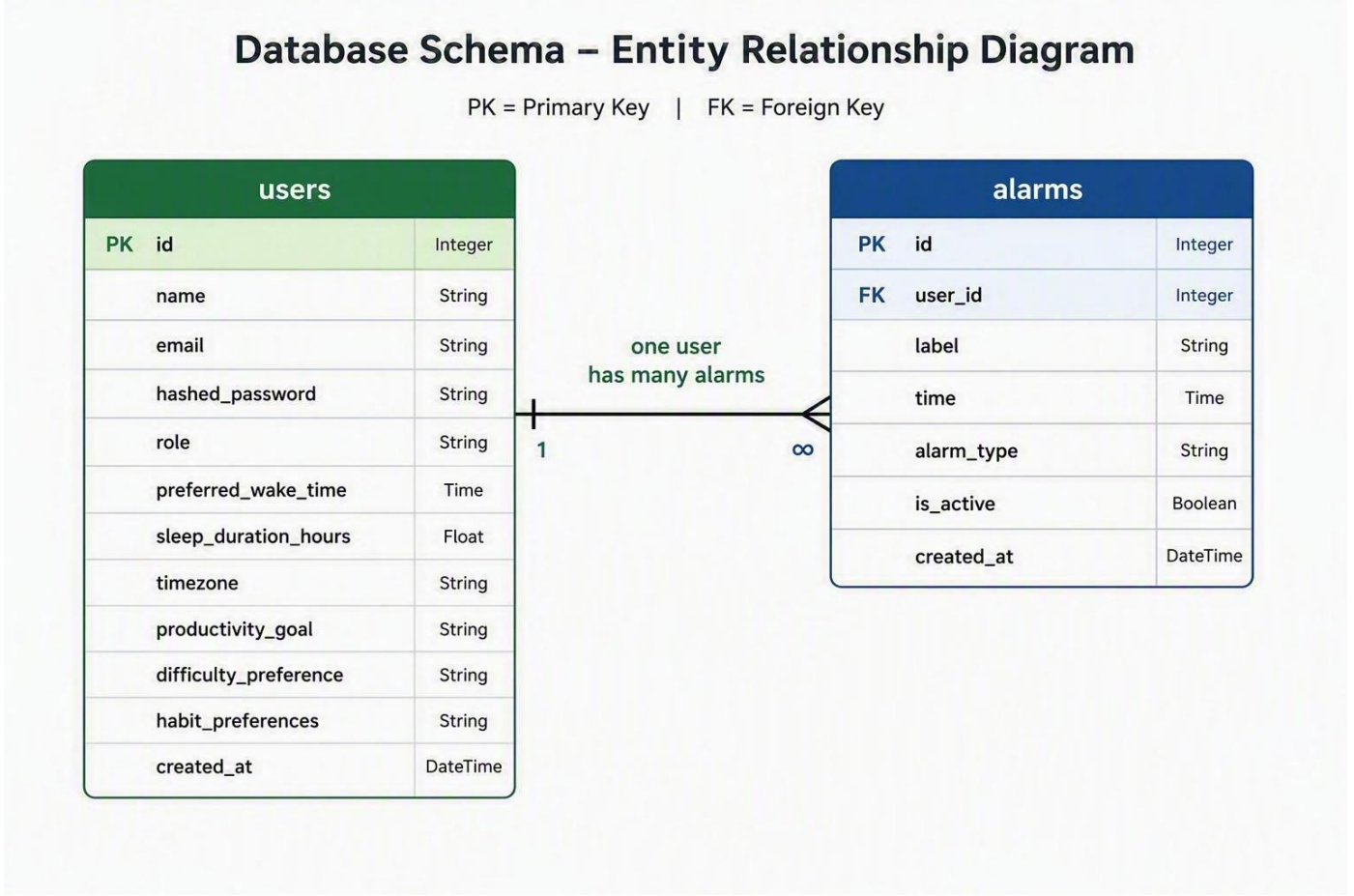
Simple layered architecture showing how the application components communicate.



DATABASE SCHEMA

Milestone 1: Initial Database Design

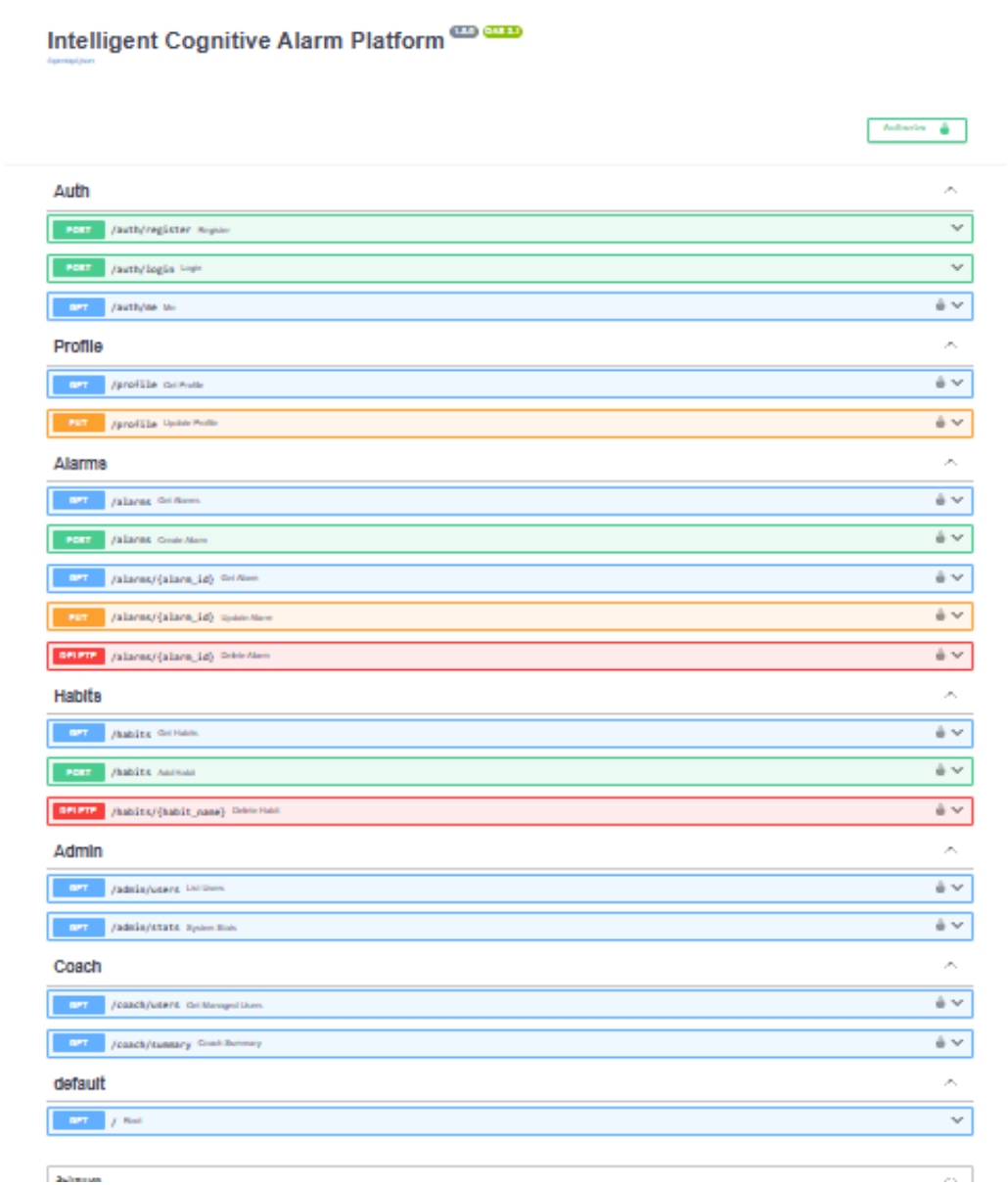
The initial relationship connects each user with one or more alarms.



BACKEND ENVIRONMENT

FastAPI

Python and FastAPI are used to expose authentication, profile and alarm APIs.



Technology stack

Frontend
- **React.js** – Used to build the user interface
- **Vite** – Frontend development/build tool
- **JavaScript** – Frontend programming language
- **React Router** – Page navigation and protected routes
- **Axios** – Connects frontend with backend APIs
- **CSS / Tailwind CSS** – UI styling

Backend
- **Python** – Backend programming language
- **FastAPI** – REST API framework
- **Uvicorn** – Runs the FastAPI server
- **SQLAlchemy** – ORM for database interaction
- **Pydantic** – Request/response validation

Database
- **SQLite** – Used as the development database for Milestone 1
- **PostgreSQL-ready architecture** – Planned/compatible for production

Authentication & Security
- **JWT (JSON Web Token)** – User authentication
- **Argon2 password hashing** – Secure password storage
- **Role-Based Access Control (RBAC)** – User, Wellness Coach, Administrator

Development Tools
- **Visual Studio Code** – Code development
- **Git/GitHub** – Version control *(only mention Git/GitHub if you actually used it)*
- **PowerShell** – Running backend/frontend commands
- **FastAPI Swagger UI** – API testing and documentation

WORKFLOW PLANNING

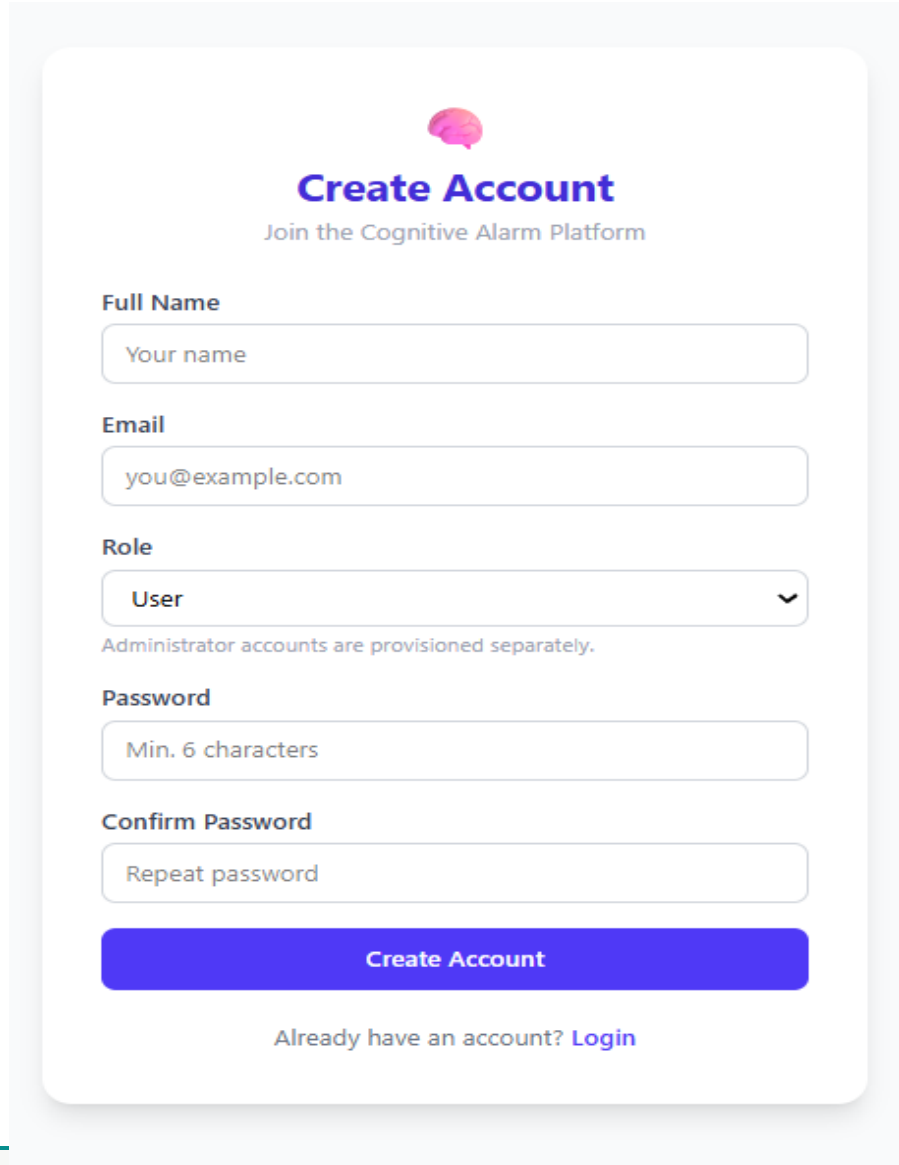
The first two milestones cover registration, alarm creation, challenge display and wake-up verification.

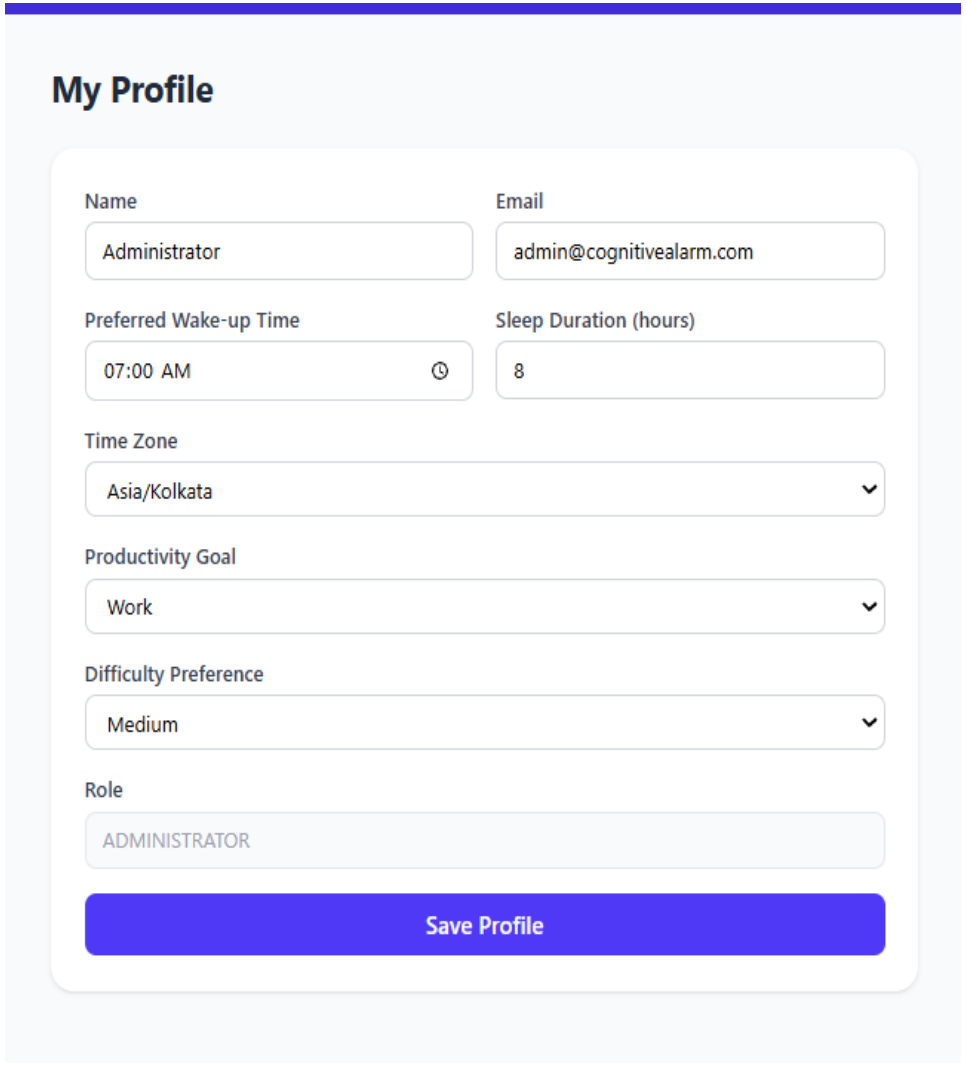


Application UI Screenshots

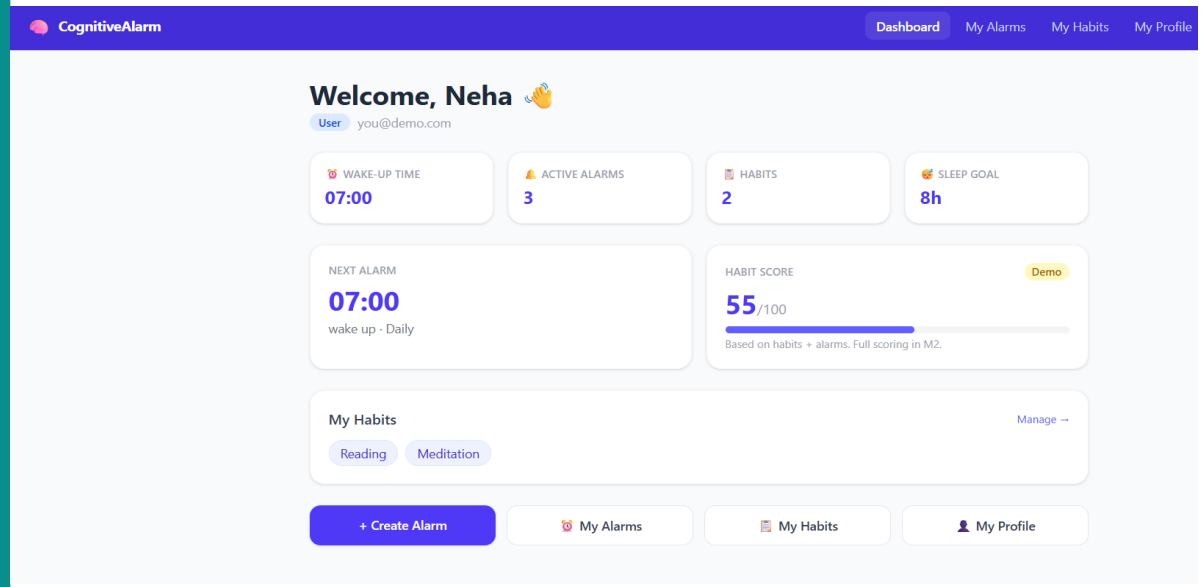
Login and Profile page

UI frames



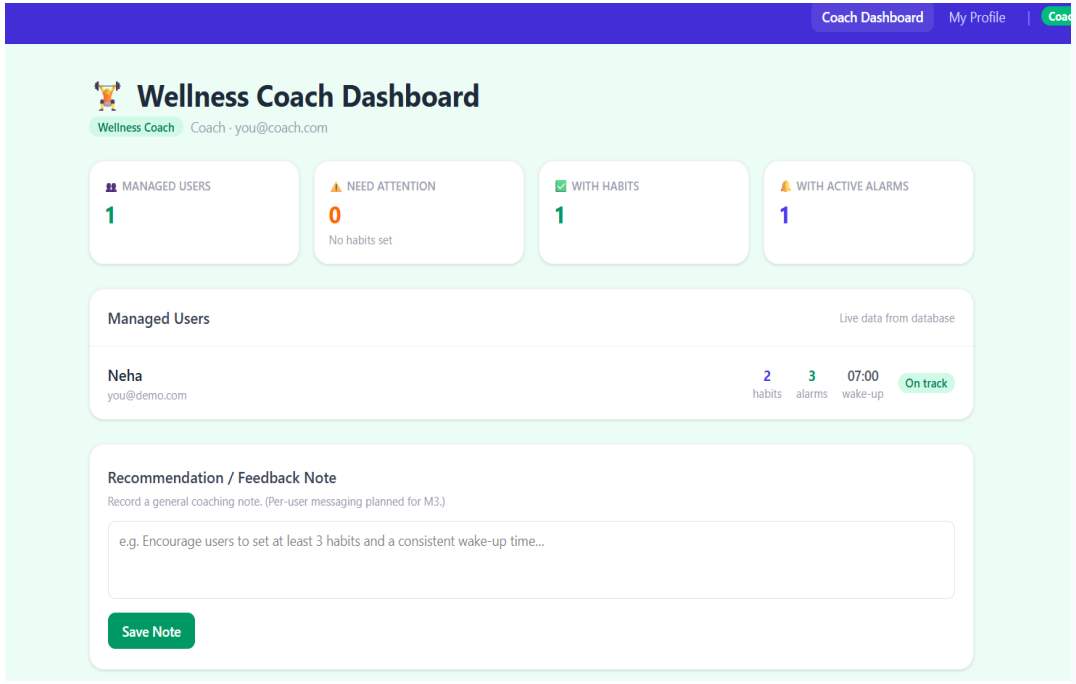


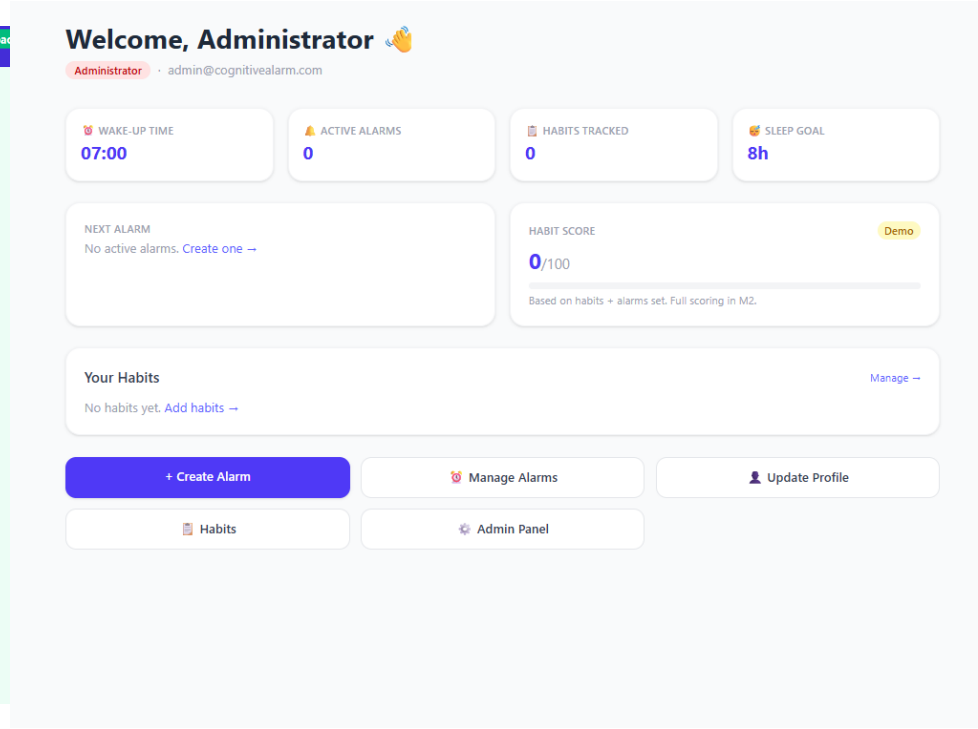
Access roles



Access roles:

- User
- Admin
- W





Summary

Summary

Developed the **basic working foundation** of the Intelligent Cognitive Alarm Platform.
Implemented **user authentication, role-based access, profiles, and alarm management**.
Connected the **React frontend with FastAPI backend** using REST APIs.
Designed the database using **SQLAlchemy with SQLite** for the current prototype.

- The platform is prepared for the next phase of **cognitive challenges, wake-up verification, and adaptive difficulty**.

Overall: Milestone 1 establishes the core platform required to build the intelligent alarm system in the upcoming phases.

Expected demo: Register → Log in → Create alarm → Alarm rings → Solve challenge → Dismiss alarm

Thank you